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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: 7th **Section:** A

Project Group ID: CSDMP230008

KRISHISAHAY – GENERATIVE AI–POWERED AGRICULTURAL QUERY RESOLUTION SYSTEM

Abstract:

Agriculture plays a vital role in supporting livelihoods and economic development, but farmers frequently face challenges such as unpredictable weather, crop diseases, pest infestations, low yields, and limited access to reliable agricultural information. Farmers often depend on multiple sources or manual consultation to make important farming decisions. Therefore, this project, "KrishiSahay – Generative AI–Powered Agricultural Query Resolution System," aims to provide a unified, intelligent, and personalized platform for addressing agricultural queries and supporting informed decision-making. The project is motivated by the need to simplify access to agricultural knowledge and provide timely recommendations based on farmers' specific conditions, location, language, and requirements. The proposed methodology involves collecting information, integrating agricultural knowledge with real-time weather and market data, processing user queries using Generative AI and Natural Language Processing, and analysing uploaded crop images using multimodal AI techniques. The system is designed to understand natural language queries and provide relevant, easy-to-understand responses that can assist farmers in identifying problems and selecting appropriate agricultural practices. It can also combine information from different sources to provide more context-aware and reliable recommendations for crop management, disease identification, weather-related decisions, and market awareness. The project also contributes to SDG 2: Zero Hunger by supporting agricultural productivity and sustainable farming, SDG 8: Decent Work and Economic Growth by supporting farmers' economic well-being, and SDG 9: Industry, Innovation and Infrastructure by applying Generative AI to modernize agricultural services.

Keywords: Generative AI, Natural Language Processing (NLP), Multimodal AI, Agricultural Intelligence, Crop Disease Detection, Smart Farming.

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Mr. Moizuddin.

Signature: _____ **Date**: _____

Coordinator **Head of the Department**